

Numerical Analysis chapter 1 theoretical homework

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I

I-question

Consider the bisection method starting with the initial interval $[1.5, 3.5]$. In the following questions “the interval” refers to the bisection interval whose width changes across different loops.

- What is the width of the interval at the n th step?
- What is the supremum of the distance between the root r and the midpoint of the interval?

I-a

If the initial width of the interval is w , then the width of the interval at the n th step will be $w_n = \frac{w}{2^n} = \frac{1}{2^{n-1}}$.

I-b

This is determined by the initial width of the interval and is equal to $\frac{3.5-1.5}{2} = 1$

at the very beginning, which is 1. This value will decrease as the bisection method narrows down the interval.

II

II-question

In using the bisection algorithm with its initial interval as $[a_0, b_0]$ with $a_0 > 0$, we want to determine the root with its *relative* error no greater than ϵ . Prove that this goal of accuracy is guaranteed by the following choice of the number of steps,

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log(a_0)}{\log 2} - 1$$

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proof

As we all know, the *relative* error $\eta = \frac{|r-c|}{|r|}$, where r is the true root, and c is the approximate root found by the algorithm. The relative error being no greater than ϵ means that $\eta \leq \epsilon$, which equals to $|r - c| \leq \epsilon|r|$. Also, we have already learnt that the width of the interval at the n th step is $\frac{b_0 - a_0}{2^{n+1}}$, where n is the number of steps. Therefore, we have $\frac{b_0 - a_0}{2^{n+1}} \leq |r - c| \leq \epsilon|r|$. Assuming $|r| = a_0$ (Since a_0 is a reasonable initial estimate), we have:

$$\frac{b_0 - a_0}{2^{n+1}} \leq |r - c| \leq \epsilon a_0$$
$$\frac{b_0 - a_0}{2^{n+1}} \leq \epsilon a_0$$

So we have:

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log(a_0)}{\log 2} - 1$$

The proof is completed. $\square$

III

III-question

Perform four iterations of Newton's method for the polynomial equation $p(x) = 4x^3 - 2x^2 + 3 = 0$ with the starting point $x_0 = -1$. Use a hand calculator and organize results of the iterations in a table.

III-answer

The derivative of $p(x)$ is $p'(x) = 12x^2 - 4x$. The Newton's method iteration formula is:

$$x_{n+1} = x_n - \frac{p(x_n)}{p'(x_n)}$$

The answer was generated by *Python* which helped by AI. [1]

Iteration	Value of x_n
x_1	-0.8125
x_2	-0.770804
x_3	-0.768832
x_4	-0.768828

Table 1: Iterations of x_n

IV

IV-question

Consider a variation of Newton's method in which only the derivative at x_0 is used,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}$$

Find C and s such that

$$e_{n+1} = C(e_n)^s$$

where e_n is the error of Newton's method at step n , s is a constant, and C may depend on x_n , the true solution α , and the derivative of the function f .

IV-answer

- *Taylor Expansion*

$$f(x) = f(\alpha) + f'(\alpha)(x - \alpha) + \frac{f''(\alpha)}{2!}(x - \alpha)^2 + \dots$$

Since $f(\alpha) = 0$, we have:

$$f(x_n) = f'(\alpha)(x_n - \alpha) + \frac{f''(\alpha)}{2}(x_n - \alpha)^2 + \dots$$

$$f(x_n) \approx f'(\alpha)(x_n - \alpha) + \frac{f''(\alpha)}{2}(x_n - \alpha)^2$$

- *Error in Iteration Formula*

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{f'(\alpha)(x_n - \alpha) + \frac{f''(\alpha)}{2}(x_n - \alpha)^2}{f'(x_n)}$$

$$x_{n+1} - \alpha = x_n - \alpha - \frac{f'(\alpha)(x_n - \alpha)}{f'(x_n)}$$

$$e_{n+1} = e_n - \frac{f'(\alpha)e_n}{f'(x_n)}$$

- *calculate*

$$e_{n+1} = \left(1 - \frac{f'(\alpha)}{f'(x_n)}\right)e_n$$

So, we could know that $C = 1 - \frac{f'(\alpha)}{f'(x_0)}$ and $s = 1$

V

V-question

Within $(-\frac{\pi}{2}, \frac{\pi}{2})$, will the iteration $x_{n+1} = \tan^{-1}x_n$ converge?

V-answer

First, we note that the function $\arctan(x)$ is the inverse tangent function, and its derivative is given by:

$$\frac{d}{dx} \arctan(x) = \frac{1}{1+x^2}$$

This derivative is always less than 1 for all x in the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$, i.e.,

$$\left| \frac{1}{1+x^2} \right| < 1$$

According to *Contraction Mapping Theorem* we could learn that $x_{n+1} = \tan^{-1}x_n$ is converge and the fixed point is $(0, 0)$.

VI

VI-question

Let $p > 1$. What is the value of the following continued fraction?

$$x = \frac{1}{p + \frac{1}{p + \frac{1}{p + \dots}}}$$

Prove that the sequence of values converges. (Hint: this can be interpreted as $x = \lim_{n \rightarrow \infty} x_n$, where

$$x_1 = \frac{1}{p}, \quad x_2 = \frac{1}{p + \frac{1}{p}}, \quad x_3 = \frac{1}{p + \frac{1}{p + \frac{1}{p}}}, \quad \dots, \quad \text{and so forth.}$$

Formulate x as a fixed point of some function.)

VI-answer

This continued fraction can be written as: $f(x) = \frac{1}{p+x}$; $x = f(x)$, which is a fixed point of the function $f(x)$. It is obvious that $\forall i, x_i > 0$. The derivative of $f(x)$ is:

$$f'(x) = -(p+x)^{-2} < 0$$

Since $p > 1, x_1 = \frac{1}{p} < 1$, so we have $x_{n+1} < x_n < 1$, which make sure that $|f'(x)| < \frac{1}{p^2} < 1$, so the sequence of values converges.

VII

VII-question

What happens in *problem II* if $a_0 < 0 < b_0$? Derive an inequality of the number of steps similar to that in *II*. In this case, is the relative error still an appropriate measure?

VII-answer

Same as *problem II*, we have: the *relative* error $\eta = \frac{|r-c|}{|r|}$, where r is the true root, and c is the approximate root found by the algorithm. We assume that α is the true root, ϵ is error, and x_n is the approximate root found by the algorithm. Also from *problem I*, we have: $x_n - \alpha \leq \frac{|b_0 - a_0|}{2^{n+1}}$, for $\eta < \epsilon$ so we have:

$$\frac{|b_0 - a_0|}{2^{n+1} * |\alpha|} \leq \epsilon$$
$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log(|\alpha|)}{\log 2} - 1$$

The relative error cannot be an appropriate measure, since the estimate of n in this expression depends on α . So, we should use the absolute error instead of the relative error.

VIII

VIII-question

(*) Consider solving $f(x) = 0$ by Newton's method with the starting point x_0 close to a root of multiplicity k . Note that α is a zero of multiplicity k of the function f if

$$f^{(k)} \neq 0; \forall i < k, f^{(i)}(\alpha) = 0$$

- How can a multiple zero be detected by examining the behavior of the points $(x_n, f(x_n))$?
- Prove that if r is a zero of multiplicity k of the function f , then quadratic convergence in Newton's iteration will be restored by making this modification:

$$x_{n+1} = x_n - k \frac{f(x_n)}{f'(x_n)}$$

VIII-answer

question1

Because α is a zero of multiplicity k of the function f , we assume that $f(x) = (x - \alpha)^k g(x)$ and $g(\alpha) \neq 0$. From the Newton's method iteration formula, we have:

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ x_{n+1} - \alpha &= x_n - \alpha - \frac{f(x_n)}{f'(x_n)} \\ x_{n+1} - \alpha &= (x_n - \alpha) - \frac{(x_n - \alpha)g(x_n)}{kg(x_n) + (x_n - \alpha)g'(x_n)} \end{aligned}$$

Now we assume $e_i = x_i - \alpha$, and we have:

$$\frac{e_{n+1}}{e_n} = 1 - \frac{g(x_n)}{kg(x_n) + e_n g'(x_n)}$$

Let's take the limit on both sides of the equation.

$$\lim_{n \rightarrow \infty} \frac{e_{n+1}}{e_n} = 1 - \frac{1}{k}$$

As we know, the convergence rate of Newton's method is quadratic, and in this case, the convergence rate is linear. So when used Newton's method, we find the convergence rate is linear, we can know that the root is a multiple zero of the function f . (This method is inspired by AI.)[\[2\]](#)

question2

similarly, we assume that $f(x) = (x - \alpha)^k g(x)$ and $g(\alpha) \neq 0$

$$\begin{aligned} x_{n+1} - \alpha &= x_n - \alpha - k \frac{(x_n - \alpha)g(x_n)}{kg(x_n) + (x_n - \alpha)g'(x_n)} \\ x_{n+1} - \alpha &= (x_n - \alpha) \left[1 - k \frac{g(x_n)}{kg(x_n) + (x_n - \alpha)g'(x_n)} \right] \\ e_{n+1} &= e_n^2 \frac{g'(x_n)}{kg(x_n) + e_n g'(x_n)} \end{aligned}$$

$$\lim_{n \rightarrow \infty} \left| \frac{e_{n+1}}{e_n^2} \right| = \lim_{n \rightarrow \infty} \left| \frac{g'(x_n)}{kg(x_n) + e_n g'(x_n)} \right|$$

And we know that $e_n \rightarrow 0$ and $x_n \rightarrow \alpha$, so we have:

$$\lim_{n \rightarrow \infty} \left| \frac{e_{n+1}}{e_n^2} \right| = \left| \frac{g'(\alpha)}{kg(\alpha)} \right|$$

and we know that $g(\alpha) \neq 0$ and $g'(\alpha)$ is a const number, so we have proved that the convergence rate is quadratic.

□

Acknowledgement

References

- [1] Moonshot AI. *Kimi Assistant*. <https://kimi.moonshot.cn/>. Accessed: 2024-09-12.
- [2] OpenAI. *ChatGPT: A Conversational Agent based on GPT-4 Architecture*. <https://chat.openai.com>. Accessed: 2024-09-13.